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Bridging Past and Future: End-to-End Autonomous Driving with Historical Prediction and Planning

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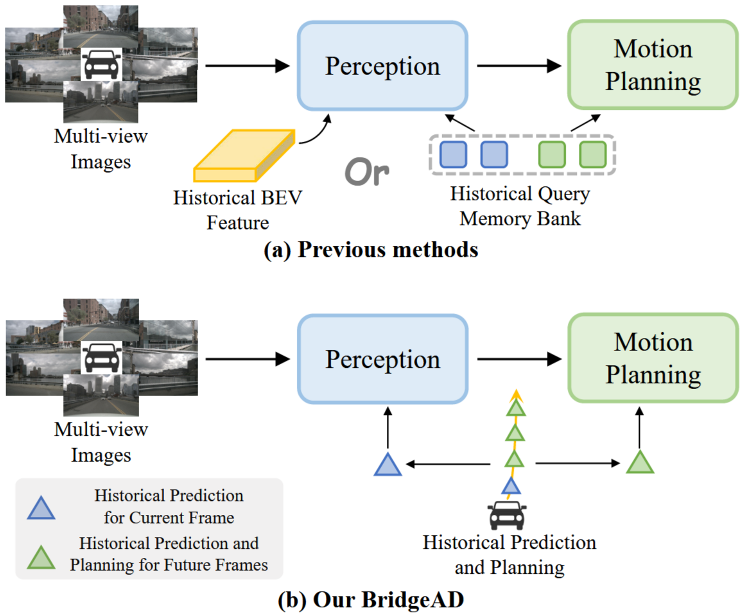
https://github.com/fudan-zvg/BridgeAD



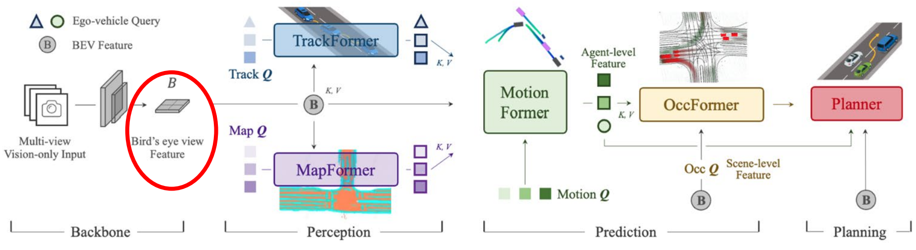
- Problem/Objective
  - End-to-End Autonomous Driving
- Contribution/Key Idea
  - Historical 정보를 통합하는 새로운 프레임워크 제안
  - “미래는 과거의 연속이다” 라는 철학에 기반
  - open-loop/closed-loop에서 둘다 SOTA

김범준

Bridging Past and Future: End-to-End Autonomous Driving with Historical Prediction and Planning

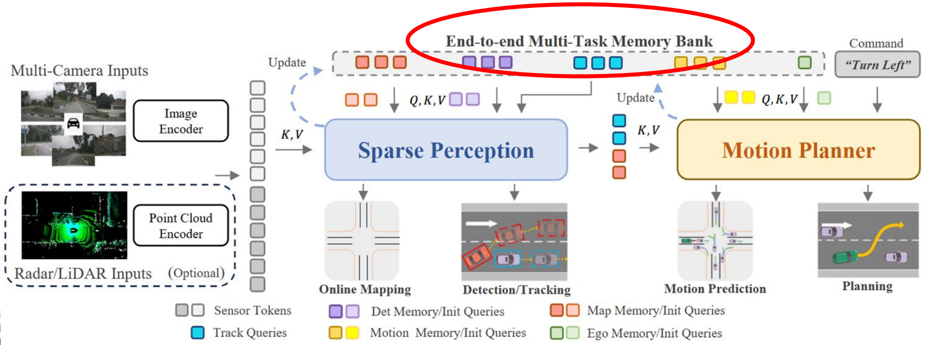
- ## Introduction

[Dense BEV feature aggregation 기반]



UniAD [1]

[Sparse memory bank 기반]



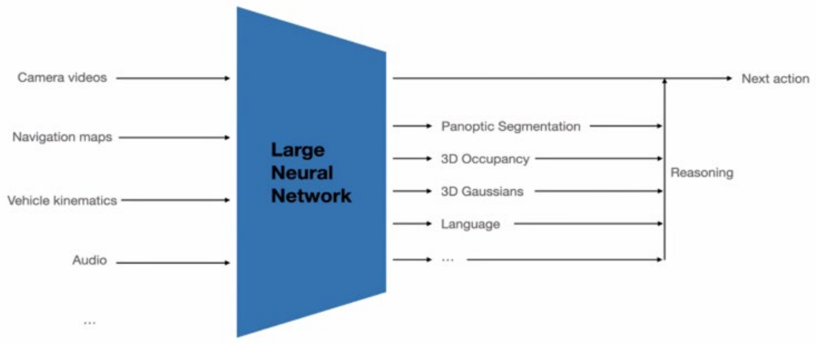
SparseAD [2]

- 기존 방법들의 한계점 분석
  - [1] BEV feature가 perception 단계에서만 사용됨 → planning, tracking에는 temporal cues 사용 x
  - [2] 하나의 query가 전체 trajectory를 대표함 → time align x, planning에는 coarse 한 temporal info
- 종합
  - perception만 temporal 정보 사용
  - Historical motion planning queries가 **step-specific reasoning**을 반영 x

[1] Hu, Yihan, et al. "Planning-oriented autonomous driving." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023.
[2] Zhang, Diankun, et al. "Sparsead: Sparse query-centric paradigm for efficient end-to-end autonomous driving." *arXiv preprint arXiv:2404.06892* (2024).

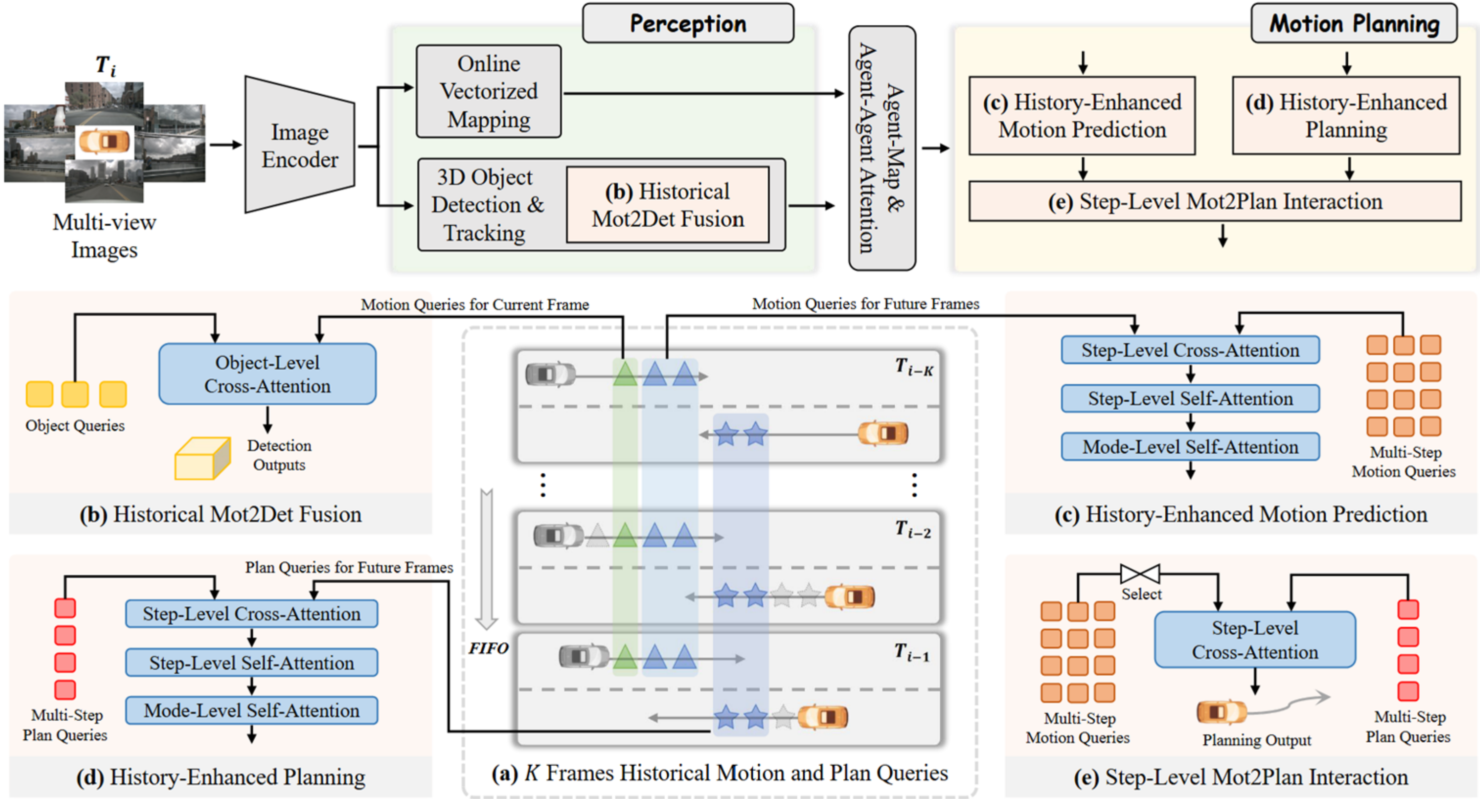
- # Introduction

[Tesla의 end-to-end Autonomous Driving]



- 각 모듈의 역할 및 존재 이유
- 전체 구조 + 각 모듈이 상호작용하는 방법

- **Framework**



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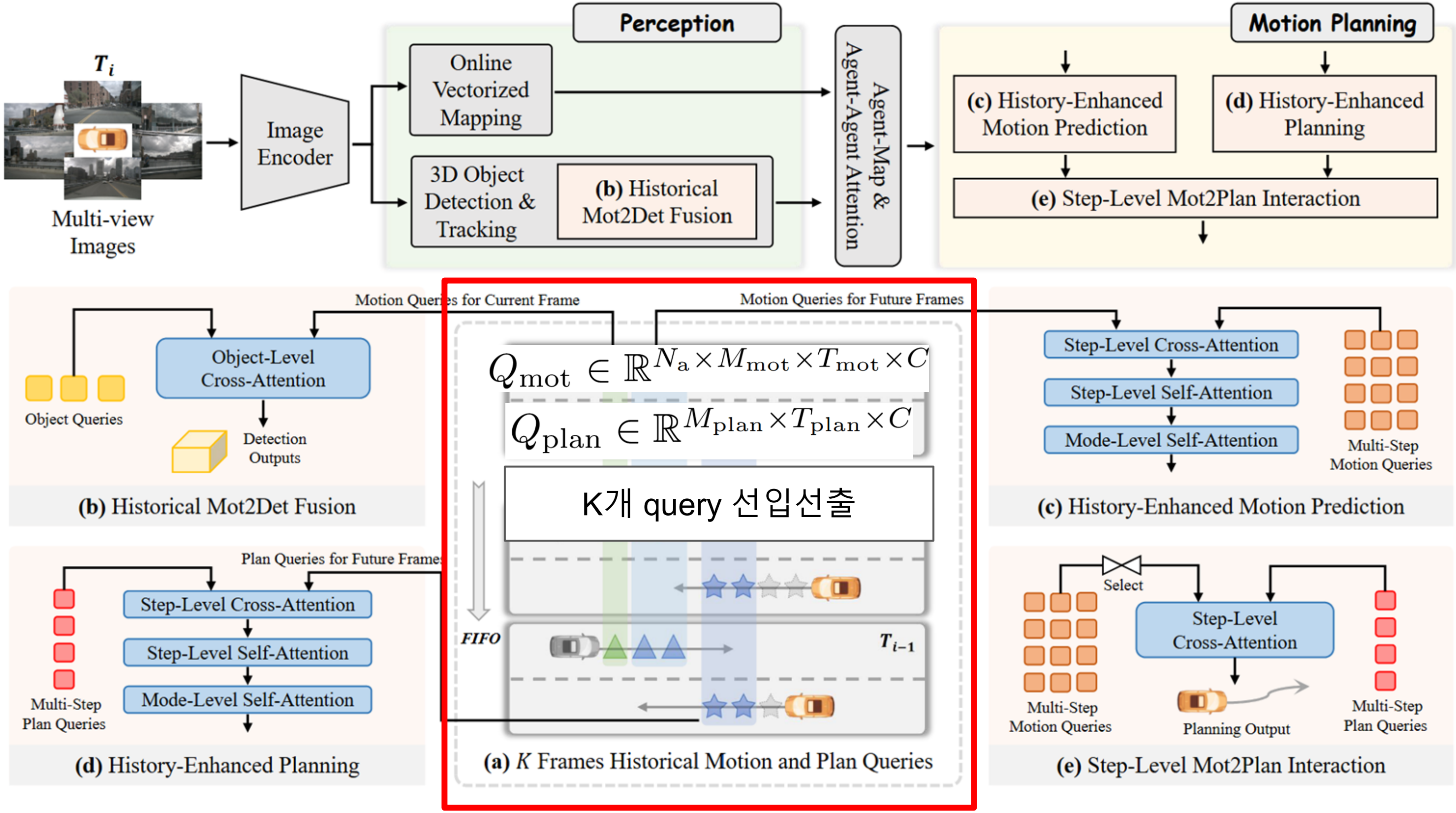
- **Framework**



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- **Contribution**

$$Q_{\text{mot}} \in \mathbb{R}^{N_{\text{a}} \times M_{\text{mot}} \times T_{\text{mot}} \times C} \longleftrightarrow Q_{\text{mot}}^{\text{previous}} \in \mathbb{R}^{N_{\text{a}} \times M_{\text{mot}} \times C}$$

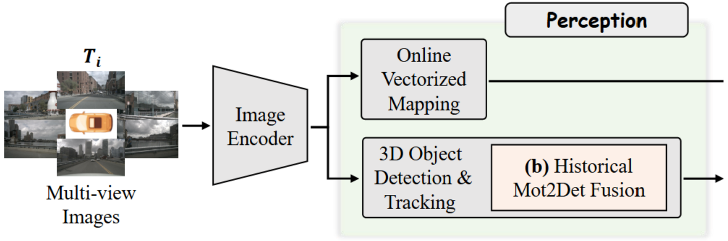
- 기존 연구
  - 한 agent x 한 mode = 하나의 trajectory 전체를 대표하는 하나의 embedding (multi-step planning과 맞지 x)
- BridgeAD
  - 모든 time step마다 별도의 query embedding (Tmot)
  - agent별 / mode 별 / future time 별 로 query가 존재
  - **step-level attention**을 적용하여 각 future step의 행동 변화(가속, 감속, 회전 등)를 fine-grained하게 reasoning 가능

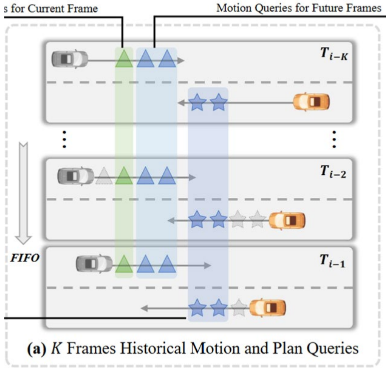
$$Q_{\text{plan}} \in \mathbb{R}^{M_{\text{plan}} \times T_{\text{plan}} \times C}$$

Method

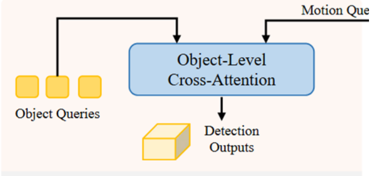
- Image encoder
  - In: 6 images / Out: Spatial feature maps (*F*)
- Perception 모듈
  - 3D OD / Tracking $B_{\mathrm{obj}} \in \mathbb{R}^{N_a \times 11}$ $Q_{\mathrm{obj}} \in \mathbb{R}^{N_a \times C}$
    - In: *F* + Obj *q* / Out: ID
  - Online vectorized mapping
    - In: *F* + map *q* / Out: map points
- (a) Memory Queue
  - K개의 motion / planning *q* cache (Qmot, Qplan)
- (b) Historical Mot2Det Fusion
  - In : (현재) obj *q* + (현재) motion *q*
  - Out: (temporal 정보 결합된) Qobj

$$Q_{\mathrm{obj}} = \mathrm{CrossAttn}(\mathrm{Q} = Q_{\mathrm{obj}}, \mathrm{K}, \mathrm{V} = Q_{\mathrm{m2d}})$$





(a) *K* Frames Historical Motion and Plan Queries



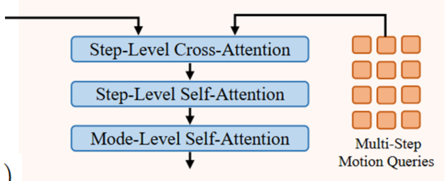
(b) Historical Mot2Det Fusion

- **Method**
  - (c) History-Enhanced Motion Prediction
    - In: multi-step motion q + (미래) motion ~
    - Out: (refined) Qmot
  - (d) History-Enhanced Planning
    - In: multi-step plan q + (미래) plan q
    - Out: (refined) Qplan
  - (e) Step-Level Mot2Plan Interaction
    - In: Qmot*, Qplan
    - Out: (final) Qplan

$$Q_{\text{mot}} \in \mathbb{R}^{N_a \times M_{\text{mot}} \times T_{\text{mot}} \times C}$$

$$Q_{\text{m2m}} \in \mathbb{R}^{N_a \times M_{\text{mot}} \times K \times T_{\text{m2m}} \times C}$$

$$Q_{\text{mot}} = \text{CrossAttn}(Q = Q_{\text{mot}}, K, V = Q_{\text{m2m}}),$$
$$Q_{\text{mot}} = \text{StepSelfAttn}(Q_{\text{mot}}),$$
$$Q_{\text{mot}} = \text{ModeSelfAttn}(Q_{\text{mot}}).$$

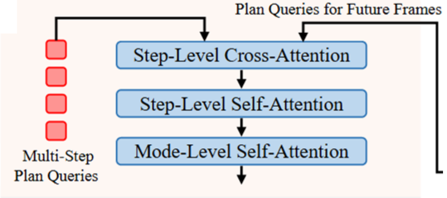


(c) History-Enhanced Motion Prediction

$$Q_{\text{plan}} \in \mathbb{R}^{M_{\text{plan}} \times T_{\text{plan}} \times C}$$

$$Q_{\text{p2p}} \in \mathbb{R}^{M_{\text{plan}} \times K \times T_{\text{p2p}} \times C}$$

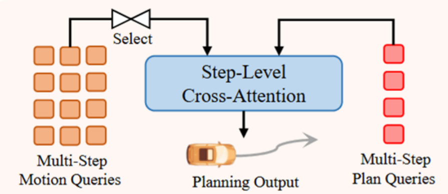
$$Q_{\text{plan}} = \text{CrossAttn}(Q = Q_{\text{plan}}, K, V = Q_{\text{p2p}})$$
$$Q_{\text{plan}} = \text{StepSelfAttn}(Q_{\text{plan}}),$$
$$Q_{\text{plan}} = \text{ModeSelfAttn}(Q_{\text{plan}}).$$



(d) History-Enhanced Planning

$$Q^*_{\text{mot}} \in \mathbb{R}^{N_a \times T_{\text{plan}} \times C}$$

$$Q^*_{\text{mot}} = \text{SelectWithScore}(Q_{\text{mot}}),$$
$$Q_{\text{plan}} = \text{CrossAttn}(Q = Q_{\text{plan}}, K, V = Q^*_{\text{mot}}).$$



(e) Step-Level Mot2Plan Interaction

- **Loss Term**

$$\mathcal{L}_{total} = \mathcal{L}_{det} + \mathcal{L}_{map} + \mathcal{L}_{mot} + \mathcal{L}_{plan}$$

$B_{\mathrm{obj}} \in \mathbb{R}^{N_{\mathrm{a}} \times 11}$ $Q_{\mathrm{obj}} \in \mathbb{R}^{N_{\mathrm{a}} \times C}$

3D boxes, classes
vs
GT boxes, classes

Predicted HD map vectors
vs
GT HD map vectors

$Q_{\mathrm{mot}} \in \mathbb{R}^{N_{\mathrm{a}} \times M_{\mathrm{mot}} \times T_{\mathrm{mot}} \times C}$

Predicted agent trajectories
vs
GT agent trajectories

$Q_{\mathrm{plan}} \in \mathbb{R}^{M_{\mathrm{plan}} \times T_{\mathrm{plan}} \times C}$

Predicted ego planning trajectory
vs
GT ego trajectory

- **Experiments**

- Open-Loop (Nuscenes)
  - 모델이 예측한 ego trajectory는 실제로 차량을 움직x
  - 단순히 GT와 "얼마나 가까운지"만 비교
  - planning quality" 측정
    - 행동의 연속성(temporal consistency)을 측정하지 못함
    - 실제 시뮬레이션 안정성 필요 없음

모델의 "순수한 예측 정확도"를 측정
GT trajectory를 얼마나 잘 따라가냐

- Closed-Loop (NeuroNCAP)
  - 현재 상태 → trajectory 예측 + 그 trajectory로 차량을 실제로 움직임
  - 모델의 output이 시스템의 다음 input이 되는 구조
    - Open-loop와 다르게 stability, consistency, temporal dependence까지 반영
    - 시뮬레이션이 필요해 비용/시간 큼

"안전성(stability)"과 "충돌 회피 능력"을 측정
Temporal feedback 존재

기존 연구들 대비 collision rate 크게 감소

- **Experiments**

Method	Reference	L2 (m) ↓				Col. Rate (%) ↓				FPS
		1s	2s	3s	Avg.	1s	2s	3s	Avg.	
OccWorld-D [64]	ECCV 2024	0.52	1.27	2.41	1.40	0.12	0.40	2.08	0.87	-
Drive-WM [52]	CVPR 2024	0.43	0.77	1.20	0.80	0.10	0.21	0.48	0.26	-
ST-P3 [18]	ECCV 2022	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	1.6
GenAD [65]	ECCV 2024	0.36	0.83	1.55	0.91	0.06	0.23	1.00	0.43	**6.7**
UniAD† [19]	CVPR 2023	0.45	0.70	1.04	0.73	0.62	0.58	0.63	0.61	1.8
VAD† [26]	ICCV 2023	0.41	0.70	1.05	0.72	0.03	0.19	0.43	0.21	4.5
SparseDrive† [47]	arXiv 2024	0.30	0.58	0.95	0.61	**0.01**	**0.05**	0.23	0.10	6.1
BridgeAD-S (Ours)	-	**0.29**	**0.57**	**0.92**	**0.59**	**0.01**	**0.05**	**0.22**	**0.09**	5.0
BridgeAD-B (Ours)	-	**0.28**	**0.55**	**0.92**	**0.58**	**0.00**	**0.04**	**0.20**	**0.08**	3.9

Table 1. **Open-loop planning results** on the nuScenes [2] validation dataset. † indicates evaluation with the official checkpoint. FPS is measured on one NVIDIA RTX 3090 GPU with batch size 1, while UniAD's is on one NVIDIA Tesla A100. To avoid the ego-status leakage problem, as proposed by Li *et al*. [29], we do not use the ego status as input.

- **Experiments**

Method	Post-proc.	Score ↑	Col. Rate (%) ↓
VAD [26]	✗	0.66	92.5
UniAD [19]	✗	0.73	88.6
SparseDrive† [47]	✗	0.92	93.9
BridgeAD-S (Ours)	✗	**1.52**	**76.2**
BridgeAD-B (Ours)	✗	**1.60**	**72.6**
VAD [26]	✓	2.75	50.7
UniAD [19]	✓	1.84	68.7
BridgeAD-S (Ours)	✓	**2.98**	**46.1**
BridgeAD-B (Ours)	✓	**3.06**	**44.3**

Table 2. **Closed-loop simulation results** on nuScenes dataset with NeuroNCAP [38] benchmark. † indicates evaluation with official checkpoint. "Post-proc." refers to trajectory post-processing, as proposed in UniAD.

- **Experiments**

Method	**ADE** $(m)\downarrow$ Car / Ped	**FDE** $(m)\downarrow$ Car / Ped	**MR** $\downarrow$ Car / Ped	**EPA** $\uparrow$ Car / Ped
ViP3D [15]	2.05 / -	2.84 / -	0.25 / -	0.23 / -
UniAD [19]	0.71 / 0.78	1.02 / 1.05	0.15 / **0.12**	0.46 / 0.35
SparseDrive [47]	**0.62** / 0.72	0.99 / 1.07	0.14 / 0.14	0.48 / 0.41
BridgeAD-S (Ours)	**0.62 / 0.70**	**0.98 / 0.99**	**0.13 / 0.13**	**0.50 / 0.44**
BridgeAD-B (Ours)	**0.60 / 0.70**	**0.96 / 0.98**	**0.13 / 0.12**	**0.52 / 0.45**

Table 3. Comparison of **motion prediction results** of state-of-the-art methods. We evaluate two main categories: cars and pedestrians.

- **Experiments**

Method	**Backbone**	**mAP ↑**	**NDS ↑**
VAD† [26]	R50	0.273	0.397
SparseDrive [47]	R50	0.418	0.525
BridgeAD-S (Ours)	R50	**0.423**	**0.534**
BEVFormer [28]	R101	0.416	0.517
UniAD [19]	R101	0.380	0.498
BridgeAD-B (Ours)	R101	**0.507**	**0.594**

(a) 3D detection results.

Method	**Backbone**	**AMOTA ↑**	**AMOTP ↓**	**IDS ↓**
ViP3D [15]	R50	0.217	1.625	-
SparseDrive [47]	R50	0.386	1.254	886
BridgeAD-S (Ours)	R50	**0.398**	**1.232**	**639**
UniAD [19]	R101	0.359	1.320	906
BridgeAD-B (Ours)	R101	**0.512**	**1.080**	**544**

(b) Multi-object tracking results.

Table 4. Comparison of **perception results** of state-of-the-art perception or end-to-end methods. † indicates evaluation with official checkpoint.

- **Experiments**

ID	HisPlan	Mot2Plan	L2 (m) ↓				Col. Rate (%) ↓			
			$1s$	$2s$	$3s$	Avg.	$1s$	$2s$	$3s$	Avg.
1		✓	0.35	0.68	1.10	0.71	0.01	0.11	0.34	0.15
2	✓		0.33	0.65	1.07	0.68	0.01	0.13	0.40	0.18
3	✓	✓	0.29	0.57	0.92	0.59	0.01	0.05	0.22	0.09

Table 5. Ablation study on the History-Enhanced Planning module and Step-Level Mot2Plan Interaction module.

ID	Mot2Det	HisMot	Detection		Tracking		Motion Prediction		
			mAP ↑	NDS ↑	AMOTA ↑	AMOTP ↓	ADE (m) ↓	FDE (m) ↓	EPA ↑
1	✓		0.412	0.526	0.387	1.240	0.66 / 0.75	1.05 / 1.08	0.47 / 0.40
2		✓	0.404	0.512	0.369	1.260	0.62 / 0.69	0.99 / 0.98	0.49 / 0.43
3	✓	✓	0.423	0.534	0.398	1.232	0.62 / 0.70	0.98 / 0.99	0.50 / 0.44

Table 6. Ablation study on the Historical Mot2Det Fusion module and History-Enhanced Motion Prediction module. We evaluate motion prediction for cars and pedestrians.

- **Experiments**

HisMot	HisPlan	Avg. L2 (m) ↓	Avg. Col. Rate (%) ↓
5	3	0.63	0.13
7	3	0.62	0.09
6	2	0.64	0.13
6	4	0.60	0.11
6	3	0.59	0.09

Table 8. Ablation study on the number of time steps for aggregating historical information.

$$Q_{\text{mot}} \in \mathbb{R}^{N_{\text{a}} \times M_{\text{mot}} \times T_{\text{mot}} \times C}$$

$$Q_{\text{plan}} \in \mathbb{R}^{M_{\text{plan}} \times T_{\text{plan}} \times C}$$

- **Experiments**

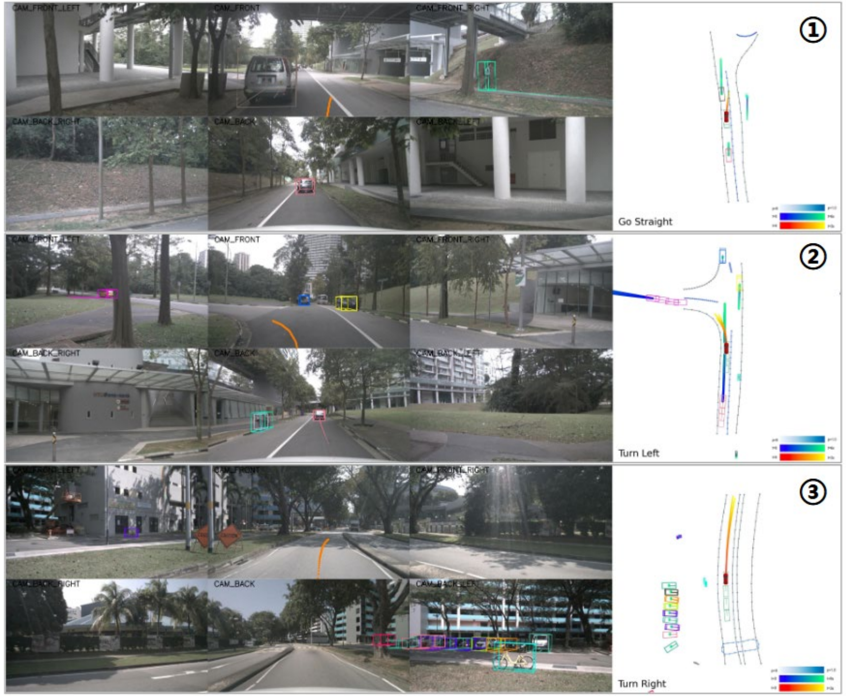


Figure 3. Qualitative results in the open-loop evaluation show that our BridgeAD accurately produces planning outputs.

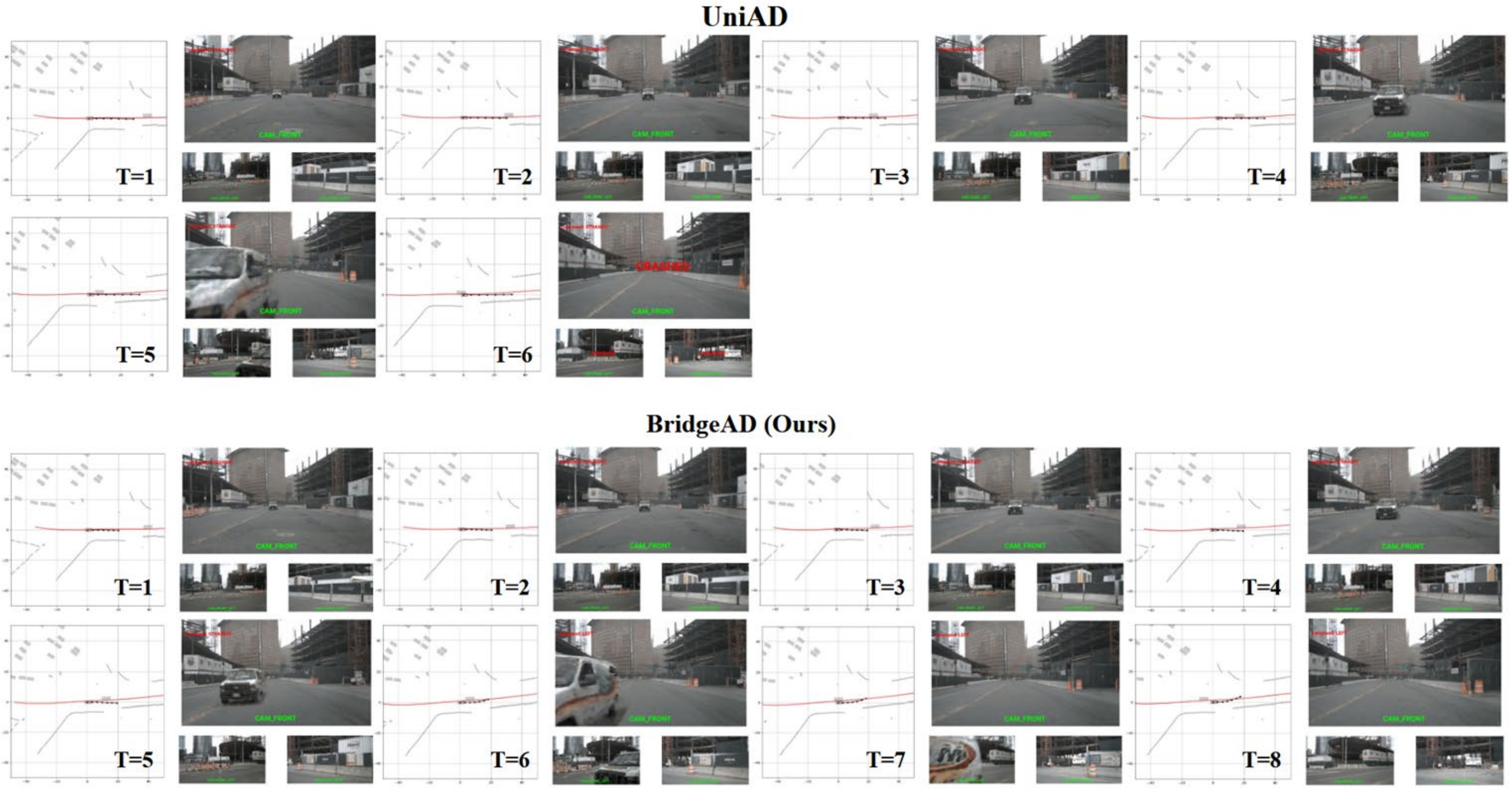


Figure 4. Qualitative results in the closed-loop evaluation demonstrate that our BridgeAD effectively avoids collisions in safety-critical scenarios.